

Data Structures Project for Fall24

NU Management System (NUMS)

1. Overview

The project is a management system in the terminal for managing the data of enrolled students, available courses and more. It uses header files to divide the tasks and text files to store data.

2. Features

- Adding, removing, finding, sorting, and displaying students
 - Adding, removing, finding, sorting, and displaying courses
 - Enrolling students in courses
 - Dropping students from courses
 - Managing course prerequisites
 - Handling course waitlists
 - Updating student data and contact information
 - Saving and loading data from files
 - Keeping track of operations using a log file
 - Using multiple text and header files to organize the project
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3. Project Structure

3.1 Main program (Main.cpp)

Which uses the ManagementSystem.h header file and contains the main function that loads the data from text files.

3.2 Header files

They were designed to divide the project into multiple parts (divide and conquer technique).

- ***ManagementSystem.h***

Contains all the fundamental functions and includes other header files.

- ***SLL.h***

Contains the necessary classes for the single linked list of the students' records.

- ***DLL.h***

Contains the double linked list of the enrolled courses. Each student has their own list.

- ***BST.h***

Contains the binary search tree that stores the data of each available course in the university.

- ***Stack.h***

Contains the stack data structure and the function of validating prerequisites.

- ***Queue.h***

Contains the queue data structure that is responsible for handling waiting lists.

- ***FileHandling.h***

This is responsible for handling the operations related to text files like loading and saving data.

- ***Log.h***

Responsible for the log file that stores all the operations done during the use of the system.

- ***SortOperations.h***

Responsible for the sorting algorithms in the projects that are basically based on quick sort algorithm to sort the single linked list by names and IDs.

- ***SearchOptimization.h***

Contains searching algorithms that with linear search (in the SLL) and binary search (in the BST).

- ***HashTable.h***

Contains the fundamental components of the hash table.

3.3 Text files

They were made to store the data of the management system like the enrolled students, available courses, enrollment history of the students, waitlists, and more.

- **Students.txt**

Contains all the main data of students like name, ID, email, phone number, address, and password.

- **StudentsGPA.txt**

Contains the total GPA of all students.

- **Grades.txt**

Contains the final grades of all the courses out of 100 of each student.

- **EnrollmentHistory.txt**

Contains the enrollment history of each student including the name, ID, and grade of each course.

- **Courses.txt**

Contains the data of each course, including name, ID, credits, instructor, and remaining seats.

- **Prerequisites.txt**

Shows the prerequisites of each course in the university.

- **RegisteredStudents.txt**

Contains the students enrolled in each course.

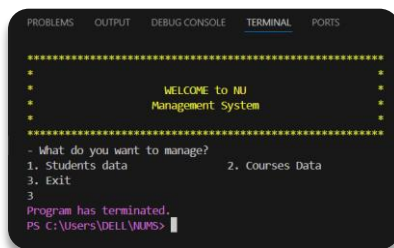
- **Waitlist.txt**

Shows the students in the waitlist of each course.

- **Log.txt**

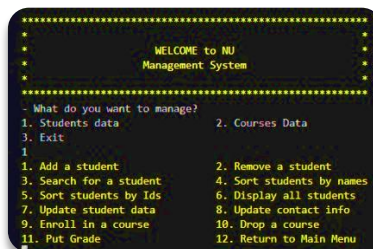
Stores all the operations done during the use of the management system.

4. The Project Visualized



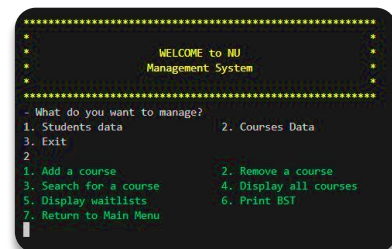
* Color Full Terminal

* Clear Organization



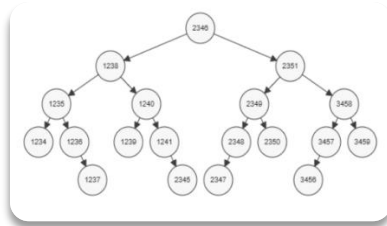
* Students' data menu

* Variety of operations

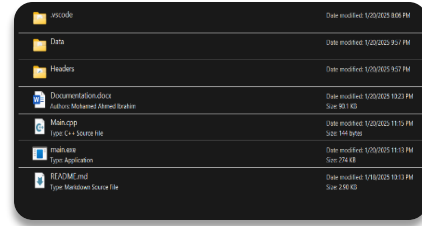


* Courses' data menu

* Suitable options to manipulate data



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Log
File Edit View
[2025-03-18 19:41:51] Grades saved to file
[2025-03-18 19:41:25] Enrolling in a course
[2025-03-18 19:41:27] Found course with ID: 1234 in hash table
[2025-03-18 19:41:29] Added course to enrolment history: Computer Skills (ID: 1234)
[2025-03-18 19:41:29] A student was registered in a course
[2025-03-18 19:41:29] Found course with ID: 1234 in hash table
[2025-03-18 19:41:29] Added course to enrolment history: Computer Skills (ID: 1234)
[2025-03-18 19:41:29] Courses and prerequisites saved to file
[2025-03-18 19:41:29] Students saved to file
[2025-03-18 19:41:29] Waitlist saved to file
[2025-03-18 19:41:29] Enrolment history saved to file
[2025-03-18 19:41:29] Registered students saved to file
[2025-03-18 19:41:29] GPA data saved to file
[2025-03-18 19:41:29] Grades saved to file
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* Balanced binary search Tree

* Log file to keep

* Clear organization and

* Low complexity operations

track of operations

division of project files

5. Team Members

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